Infosys_HotelRevAI – AI Driven Revenue Analysis for Hotels

MILESTONE – 3

DAX QUERIES AND PERFORMANCE METRICS

1.Introduction

The **Infosys_HotelRevAI – AI Driven Revenue Analysis for Hotels** project focuses on analyzing hotel booking and revenue data using advanced analytical techniques in Microsoft Power BI.

In this milestone, DAX (Data Analysis Expressions) and business performance metrics were implemented to transform raw transactional data into meaningful insights.

Business metrics define what needs to be measured in the hotel industry, while DAX queries provide the logic and computational framework required to dynamically calculate those metrics.

This section provides:

- Detailed explanation of key hotel performance metrics
- Mathematical formulas used
- Corresponding DAX implementations
- Analytical significance of each metric
- Importance of DAX in dashboard interactivity

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2. Importance of Metrics in Hotel Revenue Analysis

In the hospitality industry, performance cannot be evaluated using raw numbers alone. Structured performance indicators are required to:

- Measure revenue growth
- Monitor pricing effectiveness
- Track occupancy trends
- Identify revenue leakages
- Support forecasting and planning

3. Core Business Performance Metrics

3.1 Total Revenue

Definition

Total income generated from confirmed hotel bookings during a specific time period.

Mathematical Formula:

Revenue = ADR × Length of Stay

Analytical Importance

- Primary indicator of financial performance
- Used for monthly and yearly comparisons
- Identifies peak and off-peak revenue periods
- Helps evaluate impact of pricing changes

Total Revenue serves as the foundation for most other derived KPIs.

3.2 Total Bookings

Definition

Total number of reservations recorded in the system.

Formula

Total Bookings = Count of Booking ID

Analytical Importance

- Measures demand volume
- Indicates popularity of the hotel
- Supports occupancy analysis
- Helps compare booking trends over time

3.3 Average Daily Rate (ADR)

Definition

Average revenue earned per occupied room per day.

Formula

ADR = Total Revenue / Total Rooms Sold

Analytical Importance

- Evaluates pricing efficiency

- Supports competitive pricing strategy
- Helps detect underpricing or overpricing
- Used in revenue benchmarking

ADR is a critical pricing metric in the hospitality industry.

3.4 Occupancy Rate

Definition

Percentage of total available rooms that are occupied.

Formula

Occupancy Rate = Rooms Sold / Total Available Rooms

Analytical Importance

- Indicates demand strength
- Measures utilization efficiency
- Supports capacity management
- Helps identify seasonal demand variations

3.5 Cancellation Rate

Definition

Percentage of bookings that were canceled before check-in.

Formula-

Cancellation Rate = Cancelled Bookings / Total Bookings

Analytical Importance

- Identifies revenue leakage
- Evaluates booking reliability
- Helps improve cancellation policies
- Supports customer behavior analysis

High cancellation rates negatively impact revenue stability.

3.6 Revenue Per Available Room (RevPAR)

Definition

Revenue generated per available room, regardless of occupancy status.

Formula

RevPAR = Total Revenue / Total Available Rooms

Analytical Importance

- Combines pricing and occupancy performance
- Industry-standard KPI
- Measures overall revenue efficiency
- Helps compare performance across hotels

RevPAR provides a holistic view of revenue effectiveness.

3.7 Revenue per Booking

Definition

Average revenue generated from each reservation.

Formula

Revenue per Booking = Total Revenue / Total Bookings

Analytical Importance

- Measures customer-level profitability
- Identifies high-value bookings
- Supports targeted marketing strategies

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4. DAX Implementation in Power BI

DAX (Data Analysis Expressions) is used to create calculated measures that update dynamically based on filters and slicers.

Below are the DAX measures implemented in the Infosys_HotelRevAI project.

4.1 Total Revenue

```
Total Revenue =
SUMX(
Booking,
Booking[ADR] * Booking[Length of Stay]
)
```

Explanation

- SUMX iterates through each row of the Booking table.
- Calculates revenue per booking.
- Aggregates the total revenue dynamically.

4.2 Total Bookings

```
Total Bookings =
COUNT(Booking[Booking ID])
```

4.3 Cancelled Bookings

```
Cancelled Bookings =
CALCULATE(
COUNT(Booking[Booking ID]),
Booking[Is_Canceled] = 1
)
```

Explanation

CALCULATE modifies filter context to count only canceled bookings.

4.4 Cancellation Rate

```
Cancellation Rate =
DIVIDE(
[Cancelled Bookings],
[Total Bookings],
0
)
```

Explanation

DIVIDE ensures safe mathematical operation and prevents division errors.

4.5 Average Daily Rate (ADR)

```
ADR =
DIVIDE(
[Total Revenue],
SUM(Booking[Length of Stay]),
0
)
```

4.6 Occupancy Rate

```
Occupancy Rate =
DIVIDE(
SUM(Booking[Rooms Sold]),
SUM(Hotel[Total Rooms]),
0
)
```

4.7 RevPAR

```
RevPAR =
DIVIDE(
[Total Revenue],
SUM(Hotel[Total Rooms]),
0
)
```

4.8 Revenue per Booking

```
Revenue per Booking =
DIVIDE(
[Total Revenue],
[Total Bookings],
0
)
```

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5. Time Intelligence Measures

To analyze performance over time, advanced time-based DAX calculations were implemented.

5.1 Year-To-Date Revenue

```
YTD Revenue =
TOTALYTD(
[Total Revenue],
Date[Date]
)
```

Purpose

Calculates cumulative revenue from the beginning of the year to the selected date.

5.2 Previous Year Revenue

```
Previous Year Revenue =
CALCULATE(
[Total Revenue],
SAMEPERIODLASTYEAR(Date[Date])
)
```

5.3 Revenue Growth Percentage

```
Revenue Growth % =
DIVIDE(
[Total Revenue] - [Previous Year Revenue],
[Previous Year Revenue],
0
)
```

Purpose

Measures year-over-year growth performance.

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6. Advanced Analytical DAX

6.1 High Cancellation Risk Indicator

```
High Cancellation Risk =
IF(
AVERAGE(Booking[Lead Time]) > 30,
"High Risk",
"Low Risk"
)
```

Purpose:
Identifies bookings with higher probability of cancellation.

6.2 Revenue by Room Type

```
Revenue by Room Type =
[Total Revenue]
```

Used with Room Type dimension to compare category performance.

7. Role of DAX in Dashboard Intelligence

The implementation of DAX provides:

- Dynamic calculations
- Real-time filtering capability
- Context-aware computations
- Interactive KPI updates
- Advanced revenue forecasting support

Without DAX, dashboards would show static values.
With DAX, dashboards become intelligent analytical systems.

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8. Overall Business Impact

By integrating structured metrics and DAX calculations, the Infosys_HotelRevAI project achieves:

- Improved revenue visibility
- Better pricing strategy formulation
- Accurate occupancy measurement
- Identification of revenue leakages
- Enhanced managerial decision-making

This structured analytical framework supports long-term revenue optimization in the hospitality sector.

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9. Conclusion

In the Infosys_HotelRevAI project, business metrics define the performance indicators required for hotel revenue management, while DAX queries provide the computational logic to dynamically evaluate these metrics.

The integration of both ensures accurate analysis, real-time reporting, and strategic decision support.

This milestone successfully transforms raw booking data into actionable business intelligence using advanced analytical techniques in Power BI.